Unit circle

1 Calculate the fraction of the circumference of the unit circle the following angle measures represent:

a $x = 60^\circ$ b $x = 180^\circ$ c $s = 450^\circ$

2 State the quadrant in which the following angles are located:

a 299° b 5° c 160° d 229°

e 40° f 310° g -138° h -244°

3 Write four different angles between 0° and 360° inclusive, that lie on the quadrant boundaries.

4 For each of the following circular functions:

i State the quadrant(s) in which the function is positive.

ii State the quadrant(s) in which the function is negative.

a Sine   b Cosine   c Tangent

5 State the quadrant where the angle in each scenario is located:

a θ is an angle such that $\sin\theta > 0$ and $\cos\theta < 0$.

b θ is an angle such that $\tan\theta < 0$ and $\sin\theta > 0$.

c θ is an angle such that $\tan\theta < 0$ and $\cos\theta < 0$.

d θ is an angle such that $\tan\theta > 0$ and $\sin\theta > 0$.

6 State whether the values of the following are positive or negative:

a $\sin 31^\circ$ b $\tan 31^\circ$ c $\cos 267^\circ$ d $\sin 267^\circ$

e $\cos 180^\circ$ f $\tan 296^\circ$ g $\sin 120^\circ$ h $\cos 91^\circ$

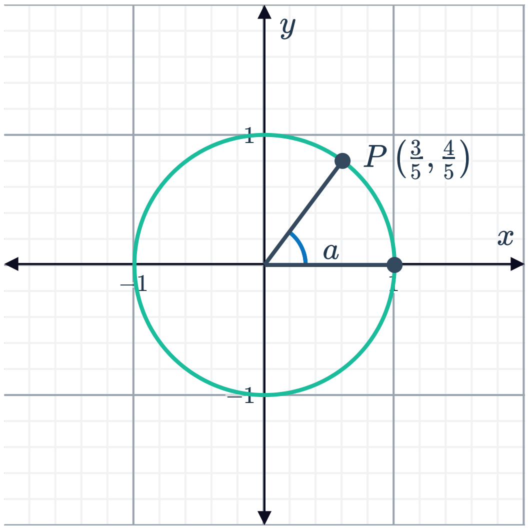
i $\sin 296^\circ$ j $\cos 120^\circ$ k $\cos 296^\circ$ l $\sin 90^\circ$

m $\cos 51^\circ$ n $\sin 51^\circ$ o $\cos 233^\circ$ p $\tan 233^\circ$

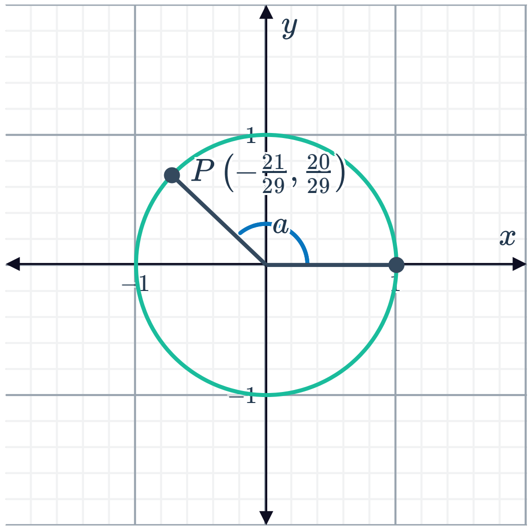
7 For each of the following graphs, state:

i $\sin a$ **ii** $\cos a$ **iii** $\tan a$

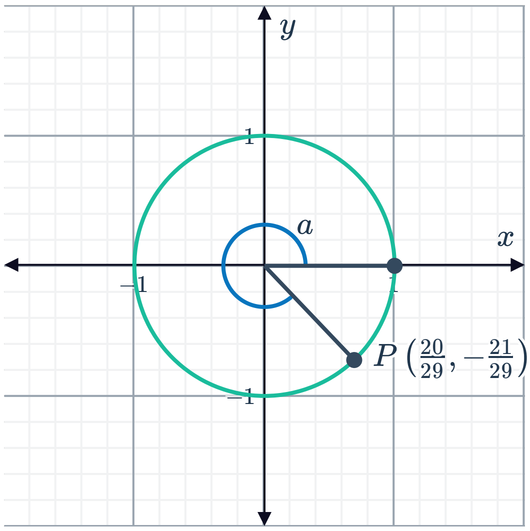
a



b



c



8 Determine whether the following statements are possible:

a $\sin\theta = \frac{7}{8}$ **b** $\sin\theta = -0.6$ **c** $\sin\theta = 1.1$ **d** $\sin\theta = -\frac{2}{3}$

9 Consider the following angles:

$$w = 253^\circ, \quad x = 265^\circ, \quad y = 193^\circ, \quad z = -258^\circ, \quad a = -182^\circ, \quad b = -257^\circ$$

Determine whether the following statements are true or false:

a Angles z, a, and b are in Quadrant 3.

b $\sin z$ and $\sin b$ are both negative.

c $\tan(z - 90^\circ)$ and $\tan(z + 90^\circ)$ are both positive.

d The angles w, x, and y all have negative cosine values.

e Angles w, x, and y are in Quadrant 3.

10 Consider the following angles:

$$w = 65^\circ, \quad x = 10^\circ, \quad y = 44^\circ, \quad z = -31^\circ, \quad a = -10^\circ, \quad b = -29^\circ$$

Determine whether the following statements are true or false:

a $\tan z$, $\cos(90^\circ + x)$ and $\sin(180^\circ + y)$ are negative.

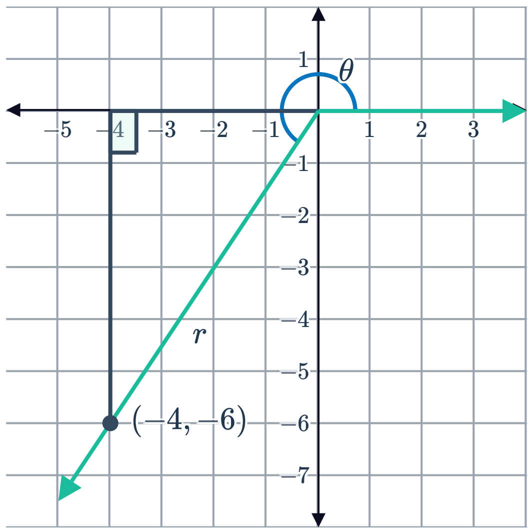
b $\sin x$, $\cos z$, and $\tan b$ are all positive.

c Angles w, x, and y are in Quadrant 1.

d $\sin x$, $\cos y$, and $\tan w$ are all positive.

e Angles z, a, and b are in Quadrant 1.

11 The terminal side of an angle, θ passes through the point $(-4, -6)$ as shown in the diagram:



a Find the value of r, the distance between $(0, 0)$ and $(-4, -6)$.

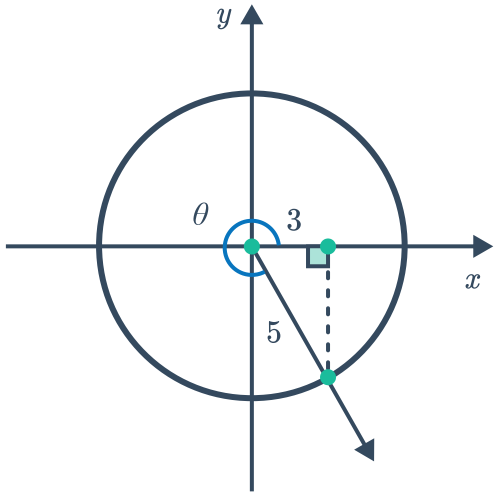
b Hence find the exact values of the following:

i $\sin\theta$ ii $\cos\theta$ iii $\tan\theta$

12 Suppose that $\cos\theta = \frac{3}{5}$, where $270^\circ < \theta < 360^\circ$. Find the exact value of the following:

a $\sin\theta$

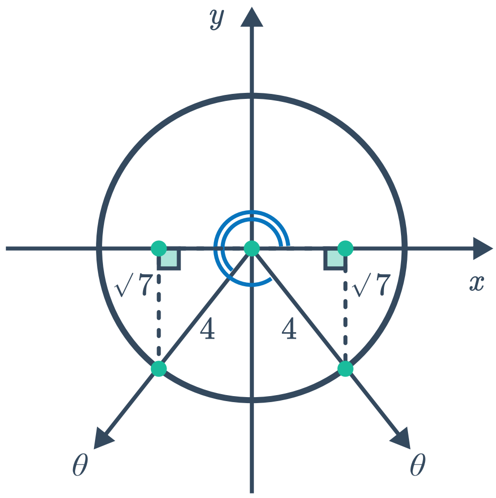
b $\tan\theta$



13 Suppose that $\sin\theta = -\frac{\sqrt{7}}{4}$. Find the exact value of the following:

a $\cos\theta$

b $\tan\theta$



14 Given angle θ such that $\sin\theta = 0.6$ and $\tan\theta < 0$, find the exact value of the following:

a $\cos\theta$

b $\tan\theta$

15 Given angle θ such that $\tan\theta = -1.3$ and $270^\circ < \theta < 360^\circ$, find the exact value of the following:

a $\cos\theta$

b $\sin\theta$

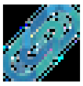
16 Given that $\tan\theta = -\frac{15}{8}$ and $\sin\theta > 0$, find the exact value of $\cos\theta$.

17 Given the following, find the exact value of $\sin\theta$.

a $\cos\theta = -\frac{60}{61}$ and $0^\circ \leq \theta \leq 180^\circ$

b $\cos\theta = -\frac{6}{7}$ and $\tan\theta < 0$

a $\cos \theta = \dfrac{3}{7}$ and θ is acute



b $\sin \theta = \dfrac{1}{\sqrt{10}}$ and $-90^\circ \leq \theta \leq 90^\circ$

Equivalent ratios

19 Consider the ratio $\sin 150^\circ$.

a State the quadrant in which 150° is located.

b State whether $\sin 150^\circ$ is positive or negative.

c Find the positive acute angle that 150° is related to.

d Hence rewrite $\sin 150^\circ$ in terms of a trigonometric ratio of a positive acute angle.

20 Consider the ratio $\cos 150^\circ$.

a State the quadrant in which 150° is located.

b State whether $\cos 150^\circ$ is positive or negative.

c Find the positive acute angle that 150° is related to.

d Hence rewrite $\cos 150^\circ$ in terms of a trigonometric ratio of a positive acute angle.

21 Consider the ratio $\tan 330^\circ$.

a State the quadrant in which 330° is located.

b State whether $\tan 330^\circ$ is positive or negative.

c Find the positive acute angle 330° is related to.

d Hence rewrite $\tan 330^\circ$ in terms of a trigonometric ratio of a positive acute angle.

22 For each of the following, rewrite the ratio as an equivalent trigonometric ratio of a positive acute angle:

a $\sin 93^\circ$

b $\cos 195^\circ$

c $\tan 299^\circ$

d $\sin 240^\circ$

e $\sin 147^\circ$

f $\tan (-50^\circ)$

g $\sin 400^\circ$

h $\cos 605^\circ$

i $\tan 525^\circ$

j $\sin 235^\circ$

k $\cos 298^\circ$

l $\tan 178^\circ$

m $\sin (-139^\circ)$

n $\cos (-222^\circ)$

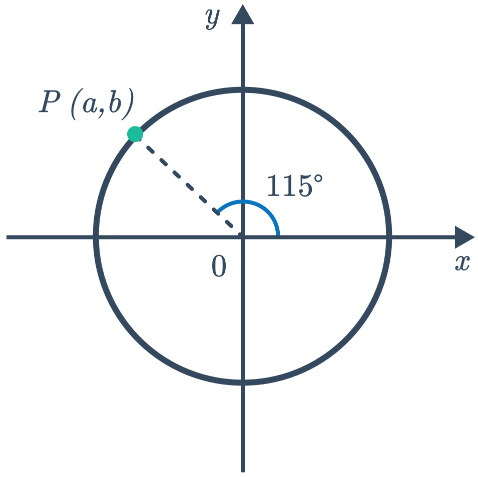
o $\tan (-289^\circ)$

p $\sin (-221^\circ)$

q $\cos (-103^\circ)$

r $\tan (-40^\circ)$

23 Consider the following diagram of a unit circle with centre O.
Express the following trigonometric ratios in terms of a and/or b:

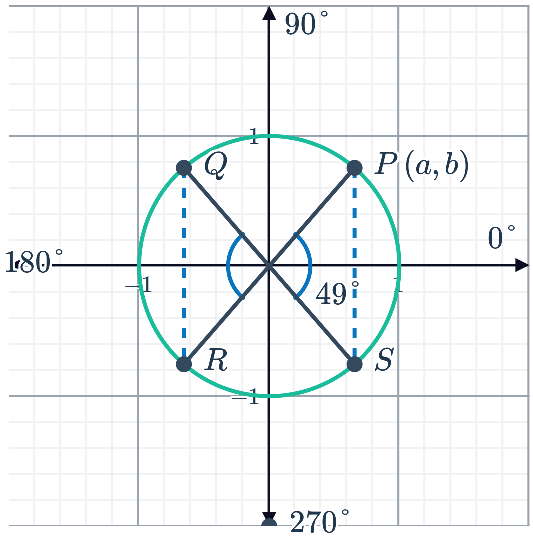


a $\tan 65^\circ$ **b** $\cos 475^\circ$

c $\sin 65^\circ$ **d** $\cos 65^\circ$

e $\cos 295^\circ$ **f** $\sin 245^\circ$

24 The diagram shows points P, Q, R and S, which represent rotations of 49°, 131°, 229° and 311° respectively around the unit circle.



a If point P has coordinates (a, b), write the coordinates of the following points in terms of a and b:

i Q **ii** R **iii** S

b State whether the following ratios are equivalent to $\sin 49^\circ$ or $-\sin 49^\circ$:

i $\sin 131^\circ$ **ii** $\sin 229^\circ$

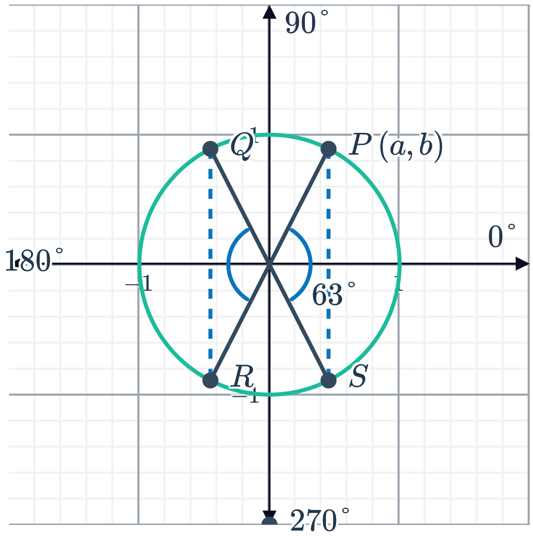
iii $\sin 311^\circ$

c Hence, express each of the following in terms of $\sin x$:

i $\sin (180^\circ - x)$ **ii** $\sin (180^\circ + x)$

iii $\sin (360^\circ - x)$

25 The diagram shows points P, Q, R and S, [illegible] represent rotations of 63°, 117°, 243° and 297° respectively around the unit circle.



a If point P has coordinates (a, b), write the coordinates of the following points in terms of a and b:

i Q ii R iii S

b State whether the following are equivalent to $-\cos 63^\circ$ or $\cos 63^\circ$:

i $\cos 117^\circ$ ii $\cos 243^\circ$

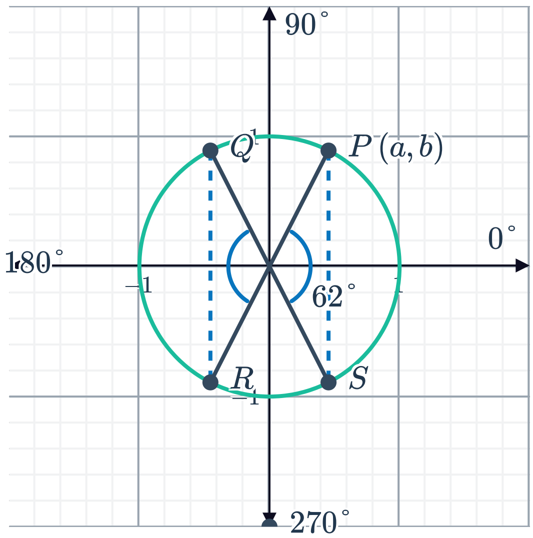
iii $\cos 297^\circ$

c Hence, express each of the following in terms of $\cos x$:

i $\cos(180^\circ - x)$ ii $\cos(180^\circ + x)$

iii $\cos(360^\circ - x)$

26 The diagram shows points P, Q, R and S, which represent rotations of 62°, 118°, 242° and 298° respectively around the unit circle.



a If point P has coordinates (a, b), write the coordinates of the following points in terms of a and b:

i Q ii R iii S

b State whether the following are equivalent to $-\tan 62^\circ$ or $\tan 62^\circ$:

i $\tan 118^\circ$ ii $\tan 242^\circ$

iii $\tan 298^\circ$

c Hence, express each of the following in terms of $\tan x$:

i $\tan(180^\circ - x)$ ii $\tan(180^\circ + x)$

iii $\tan(360^\circ - x)$

27 Simplify:

a $\sin(360^\circ + \theta)$ b $\cos(360^\circ - \theta)$ c $\sin(180^\circ - \theta)$ d $\cos(180^\circ + \theta)$

e $\sin(180^\circ + \theta)$ f $\sin(90^\circ - \theta)$ g $\sin(-\theta)$ h $\cos(-\theta)$

28 Given the approximations $\cos 21^\circ = 0.93$ a[redacted] $21^\circ = 0.36$, state the approximate values of the following, to two decimal places:

a $\cos 339^\circ$ b $\sin(-339^\circ)$ c $\cos 159^\circ$ d $\sin 159^\circ$

e $\sin 201^\circ$ f $\cos 201^\circ$ g $\sin(-201)^\circ$ h $\cos(-201)^\circ$

Use technology

29 Find the following, correct to two decimal places:

a $\sin 146^\circ$ b $\cos 126^\circ$ c $\tan 161^\circ$ d $\tan 386^\circ$

e $\cos 621^\circ$ f $\cos(-218^\circ)$ g $\sin(-42.5^\circ)$ h $-\tan(-89^\circ)$

i $(\sin 35^\circ)^2$ j $\sin^2 35^\circ$ k $\sin(35^\circ)^2$ l $-\sin^2 35^\circ$

30 Find the following:

a $\sin^2 35^\circ + \cos^2 35^\circ$ b $\sin^2(-64)^\circ + \cos^2(-64)^\circ$

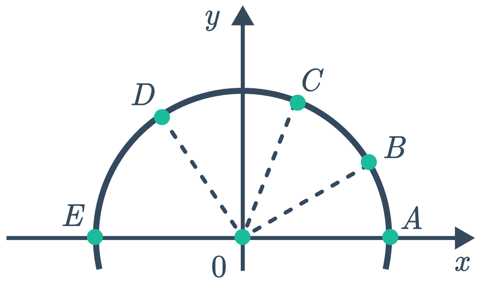
c $\sin^2 275^\circ + \cos^2 275^\circ$ d $\sin^2 x^\circ + \cos^2 x^\circ$

31 Consider the following diagram of a unit circle:

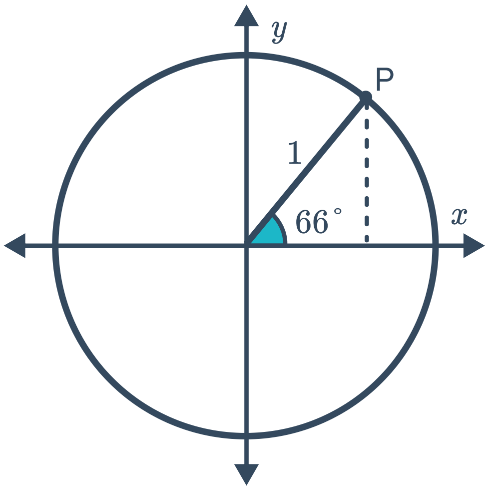
a Given that point B, represents a rotation of 35° around the unit circle, find the coordinates of B, correct to three decimal places.

b Point C has coordinates $(0.294, 0.956)$. Find the angle that point C represents to the nearest degree.

c Given that the angle subtended by arc $\angle CD = 54^\circ$, find the coordinates of D, correct to three decimal places.



32 The diagram shows P, which represents a[redacted]on of 66° around the unit circle. Find the following, rounding your answers to two decimal places:



a Coordinates of point P.

b Coordinates of point R, that represents point P reflected horizontally about the y-axis.

c Size of the rotational angle for R around the unit circle.

d Coordinates of point Q, that represents point P reflected vertically about the x-axis.

e Size of the rotational angle for Q around the unit circle.